MIL-T-10110A(Ord)
11 September 1959
SUPERSEDING
MIL-T-10110 (Ord)
30 January 1959

MILITARY SPECIFICATION

TELESCOPE, M91C

1. SCOPE

1.1 This specification covers one type of telescope known as Telescope, M91C. The telescope is used for direct fire with the 90mm Gun Tank, M46 and M46A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specifications Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-T-45070	Telescope, M91 Series and T186; Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

F7632057	Telescope, M91C

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC 1240

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3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one telescope, representative of production, shall be made available to the Government to determine its ability to meet all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Design and construction.- The telescope shall be manufactured in accordance with Drawing F7632057, "Telescope, M91C," and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired optical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specifications.- The telescope shall meet the following requirements of specifications MIL-O-13830 and MIL-F-13926.

3.4.1 MIL-O-13830

(a) Cleanliness
(b) Illuminated reticles

3.4.2 MIL-F-13926

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 Storage temperature.- The telescope shall show no evidence of physical failure and shall meet all performance requirements of this specification at room temperature (plus 60 degrees to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.5.2 Operating temperature.- The telescope shall meet the applicable requirements of 3.8.11 when exposed and thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit subsequent to which the telescope shall meet all applicable requirements to this specification at room temperature (plus 60 degrees to plus 90 degrees Fahrenheit).

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3.5.3 Vibration.- With the telescope mounted in its normal operating position, the telescope shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of five minutes plus or minus fifteen seconds. Subsequent to vibration the telescope shall not be damaged, there shall be no broken, loose or missing parts, and the telescope shall meet the applicable requirements of 3.6 to 3.8.13 inclusive.

3.6 Cleanliness.- The telescope shall meet the cleanliness requirements of specification MIL-O-13830. There shall be no more than two particles of dirt visible on the surfaces of the reticle. The area of each particle shall not be in excess of the area equivalent to a size ten dig. There shall be no dirt on the other surfaces which shall interfere with observation or adversely effect image quality when viewing through the telescope.

3.7 Sealing.- The telescope shall show no evidence of leakage in excess of 0.1 PSIG when subjected to an external pressure of 5 PSIG for a period of two minutes. The telescope shall not be immersed in water to determine area of any leak.

3.8 Performance.

3.8.1 Parallax.- Parallax shall not exceed 0.1 mil (in elevation) with respect to the horizontal line and shall not exceed 0.2 mil (laterally) with respect to the vertical line when viewing a target at a distance of 420 yards plus or minus 20 yards at the center of the field of view.

3.8.2 Eyepiece focus.- The focusing ring shall provide sufficient movement of the eyepiece to accommodate a minimum range of plus 2.5 diopters to minus 2.5 diopters.

3.8.3 Diopter setting.- The diopter setting shall not vary by more than plus or minus 0.25 diopter as the power is changed from minimum to maximum, using the minimum power as a reference.

3.8.4 Definition.- With the telescope adjusted for minimum magnification the resolving power on the optical axis for a parallel bar test target shall be 5.0 seconds of arc or better.

3.8.5 Collimation.- The intersection of the horizontal line (lateral lead reticle) and the top of the lower vertical line (ballistic reticle) shall coincide with the geometric axis of the telescope within a radial distance of 5 mils, as established by the spherical bearing surfaces of the mounting collars.

3.8.6 Reticle travel.- The zero graduation of the lateral lead scale shall not deviate from a vertical target line by more than 0.3 mil when the elevation input knob is varied between 0 and 5000 yards range (HE M71). The top of the lower vertical reticle line shall not deviate from the horizontal target line more than 0.2 mil when the lateral lead input knob is varied from zero to 20 mils right and from zero to 20 mils left.

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3.8.7 Reticle jump.

3.8.7.1 Elevation.- Reversing the lead knob shall not cause the elevation to change more than 0.2 mil.

3.8.7.2 Lead.- Reversing the elevation knob shall not cause the lead to change more than 0.2 mil.

3.8.8 Reticle accuracy.- The angular value between the zero range mark and the 5000 yard mark on the (HE M71) scale shall be 47.3 mils within 0.5 mil at any magnification.

3.8.9 Reticle illumination.- Utilizing an Instrument Light, M36, or equivalent, the reticle shall appear clearly defined when viewed in a darkened area.

3.8.10 Magnification.- The minimum magnifying power shall be not greater than 4.2 and the maximum magnifying power shall be not less than 7.6 power.

3.8.11 Torque.

3.8.11.1 Power ring.- The torque required to rotate the change of power ring from any position shall not exceed 4.5 pound inches at room temperature (plus 60 degrees to plus 90 degrees Fahrenheit) and plus 150 degrees Fahrenheit, and shall not exceed 9 pound inches at minus 40 degrees Fahrenheit.

3.8.11.2 Diopter ring.- The torque required to rotate the diopter adjustment ring shall not exceed 8 pound inches at room temperature (plus 60 degrees to plus 90 degrees Fahrenheit) and plus 150 degrees Fahrenheit, and shall not exceed 20 pound inches at minus 40 degrees Fahrenheit.

3.8.11.3 Reticle adjustment.- The torque required to rotate the azimuth and elevation reticle adjustment knobs shall not exceed 4.5 pound inches at room temperature (plus 60 degrees to plus 90 degrees Fahrenheit) and plus 150 degrees Fahrenheit, and shall not exceed 20 pound inches at minus 40 degrees Fahrenheit.

3.8.12 Knob travel (lead and elevation).- The knob stops shall allow full coverage of the lead and ballistic scales by their respective indicating lines.

3.8.13 Reticle alinement.- With the telescope horizontal with the locating pin on the rear collar vertical (upward), with the deflection scale indicating zero and the horizontal line intersecting the top of the lower vertical reticle line, the following conditions shall be met when using the minimum power position:

(a) The horizontal reticle line shall not deviate from a horizontal target line by more than the angle subtended by 0.4 mil at one extremity of the reticle line when the other extremity is in coincidence with the horizontal target line.

(b) The true vertical reticle line shall not deviate from a vertical target line by more than the angle subtended by 0.2 mil at one extremity of the reticle line when the other extremity is in coincidence with the vertical target line.

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3.9 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.

4.1.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Contractor quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the specification and other documents referenced in this specification. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that the substitute procedures will provide, as a minimum, the quality assurance required in the contractual documents. In cases of dispute as to whether certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced therein.

4.1.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and Government product inspection when deemed necessary by the Government to measure quality product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Inspection provisions.- Quality assurance provisions specified in Specifications MIL-F-13926, and MIL-O-13830 shall apply where applicable and are not in conflict with the provisions specified herein.

4.2.1 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

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4.2.2 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.2.3 Examination.- The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.2.3.1 All rejected lots shall be screened for all defective characterisitics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance inspection as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 1.5	Req't Par.	QA Par.	Test Method
1. Physical failure, post vibration	3.5.3	4.3.2	Visual-Tactile
2. Cleanliness	3.6	4.3.3	Visual
3. Eyepiece focus	3.8.2	4.3.4	Std. Measuring Equip. (SME)
4. Definition	3.8.4	4.3.5	SME
5. Magnification	3.8.10	4.3.6	SME
6. Reticle knob travel	3.8.12	4.3.7	Visual
7. Parallax	3.8.1	4.3.8.1	Device, Special Testing (DST)
8. Collimation	3.8.5	4.3.8.2	DST
9. Reticle travel	3.8.6	4.3.8.3	DST
10. Reticle jump	3.8.7	4.3.8.4	DST
11. Reticle accuracy	3.8.8	4.3.8.5	DST
12. Reticle alinement	3.8.13	4.3.8.6	DST
13. Reticle illumination	3.8.9	4.3.9	DST
14. Power ring torque	3.8.11.1	4.3.10.1	DST
15. Diopter ring torque	3.8.11.2	4.3.10.2	DST
16. Reticle adjustment	3.8.11.3	4.3.10.3	DST
17. Leak	3.7	4.3.11	DST
18. Identification marking	3.9	4.3.14	Visual

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L-7, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 4.0	Req't Par.	QA Par.	Test Method
1. Physical failure, post environmental	3.5.1 3.5.2	4.3.12.2	Visual-Tactile
2. Power ring torque (-40°F, +150°F)	3.5.2	4.3.12.3.1	DST

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MAJOR: AQL 4.0 (Cont'd)	Req't Par.	QA Par.	Test Method
3. Diopter ring torque (-40°F, ∠150°F)	3.5.2	4.3.12.3.1	DST
4. Reticle adjustment (-40°F, ∠150°F)	3.5.2	4.3.12.3.1	DST
5. Physical failure, post vibration (∠60°F to ∠ 90°F)	3.5.1 3.5.2	4.3.2 4.3.13	Visual-Tactile
6. Cleanliness	3.5.1 3.5.2	4.3.3 4.3.13	Visual
7. Eyepiece focus	3.5.1 3.5.2	4.3.4 4.3.13	SME
8. Definition	3.5.1 3.5.2	4.3.5 4.3.13	SME
9. Magnification	3.5.1 3.5.2	4.3.6 4.3.13	SME
10. Reticle knob travel	3.5.1 3.5.2	4.3.7 4.3.13	Visual
11. Parallax	3.5.1 3.5.2	4.3.8.1 4.3.13	DST
12. Collimation	3.5.1 3.5.2	4.3.8.2 4.3.13	DST
13. Reticle travel	3.5.1 3.5.2	4.3.8.3 4.3.13	DST
14. Reticle jump	3.5.1 3.5.2	4.3.13 4.3.8.4	DST
15. Reticle accuracy	3.5.1 3.5.2	4.3.8.5 4.3.13	DST
16. Reticle alinement	3.5.1 3.5.2	4.3.8.6 4.3.13	DST
17. Reticle illumination	3.5.1 3.5.2	4.3.9 4.3.13	DST
18. Power ring torque	3.5.1 3.5.2	4.3.10.1 4.3.13	DST
19. Diopter ring torque	3.5.1 3.5.2	4.3.10.2 4.3.13	DST
20. Reticle adjustment	3.5.1 3.5.2	4.3.10.3 4.3.13	DST
21. Leak	3.5.1 3.5.2	4.3.11 4.3.13	DST

4.2.4 Inspection equipment.- The test methods and procedures outlined in 4.3 shall be used as standards for determining the adequacy of the contractor's inspection equipment and methods utilized for inspecting the characteristics in Tables I and II of 4.2. Unless otherwise specified in the contract, the contractor shall provide and maintain gages and other measuring and testing devices considered necessary for inspection of supplies for conformance to contracted requirements. The Government reserves the right to certify all such equipment and to use this equipment for Government verification inspection.

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4.3 Test methods and procedures.

4.3.1 Vibration.- Each telescope in the sample shall be mounted in its normal operating position, by means of the telescope spherical mounting collars, on a device, special testing, and vibrated in accordance with the requirements specified.

4.3.2 Physical failure, post vibration.- Subsequent to vibration, the telescope shall be inspected for internal and external damage to mechanical and optical components, such as: loose, bent, broken mechanical parts, glass chip and optical cement failure. Damage shall be determined by means of a visual and tactile examination of the telescope.

4.3.3 Cleanliness, post vibration.- This test shall be conducted as specified in specification MIL-O-13830(Ord). The internal surfaces of the telescope shall show no evidence of moisture. Dirt particles shall not exceed the tolerance specified. The mean diameter of a size 10 dig is equal to 0.10 millimeter.

4.3.4 Eyepiece focus.- This test shall be performed with the aid of a calibrated dioptometer having a magnification of at least three (3x) power. Adjust the dioptometer scale to plus 2.5 diopters, the eyepiece for best focus of the dioptometer reticle, then position the dioptometer at the telescope eyepiece. When the telescope diopter ring is rotated clockwise (plus), the best focus of the telescope reticles shall be obtained before the diopter ring engages its CW stop. Adjust the dioptometer scale to minus 2.5 diopters and rotate the telescope diopter ring to the counterclockwise stop. Best focus of the telescope reticle shall be obtained before the ring engages the CCW stop.

4.3.4.1 Rotate the power ring of the telescope to minimum power and adjust the dioptometer scale to zero. While sighting through the dioptometer, positioned at the telescope eyepiece, focus the telescope diopter ring until best focus of the telescope reticle is obtained. Rotate the telescope power ring to maximum power. Any necessary adjustment of the dioptometer scale to obtain best focus of the telescope reticle shall not exceed the tolerance specified.

4.3.5 Definition.- This test shall be conducted in accordance with specification MIL-O-13830. The target image shall be viewed with the aid of a six power auxiliary telescope of approved design, scaled in diopters.

4.3.6 Magnification.- Position a diaphragm, having a 3/4 inch diameter aperature, over the telescope window so that the geometric centers of the diaphragm aperature and telescope window are approximately aligned. Adjust the telescope power ring to minimum power and place the telescope on a firm stand with the line of sight directed toward a white fluorescent light source, located approximately 12 inches from telescope window. Position a dynameter at the telescope eyelens, adjust the dynameter for best focus of the diaphragm aperature image and superimpose the dynameter reticle over this image. The diameter of the image shall not be less than 4.5 millimeters as measured on the dynameter reticle. Repeat this test with the telescope set for maximum power. The diameter of the image shall not be more than 2.5 millimeters for maximum power, as measured on the dynameter reticle.

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4.3.7 Reticle knob travel.

4.3.7.1 Rotate the azimuth reticle assembly knob to the counterclockwise (ccw) stop. Rotation of the knob in a clockwise (cw) direction shall permit the vertical reticle line to be superimposed on each graduation of the lateral lead scale before the knob engages the (cw) stop.

4.3.7.2 Rotate the elevation reticle assembly knob to the counterclockwise (ccw) stop. Rotation of the knob in a clockwise (cw) direction shall permit the horizontal reticle line to be superimposed on each range graduation, boresight point or zero range to maximum range of the ballistic scales before the knob engages the (cw) stop.

4.3.8 Positioning of telescope for parallax, collimation and reticle tests.- The tests for parallax, collimation and reticle tests shall be conducted with the telescope mounted horizontal in a device, special testing by means of the spherical bearing surfaces of the mounting collars. The verticality of the locating pin, as established by the device, special testing on the rear mounting collar, must be maintained during all reticle tests. A line of sight through the telescope shall be directed to a collimator target appropriately calibrated for conducting parallax, collimation and reticle tests. The focus of the collimator target shall be adjusted to simulate the target distance specified in 3.8.1. All reticle movements shall be accomplished by means of the appropriate reticle assembly knob.

4.3.8.1 Parallax.- Superimpose the vertical reticle line on the zero lead line of the lateral lead scale. Superimpose the horizontal reticle line on the boresight point (see Reticle Dwg D7632053), of the ballistic reticle. Direct a line of sight through the telescope boresight point to a sharply defined target object located at the specified distance from the telescope objective lens. Parallax will be recognized as any apparent movement of a reticle line in relation to the target image when the observer's head is moved side-to-side and up-and-down. Parallax at the boresight point with respect to the target image shall not exceed the tolerances specified.

NOTE: 0.1 MIL = Thickness of a reticle line.

4.3.8.2 Collimation.- Position the lines of the reticles as specified in 4.3.8.1. The radial distance, as measured on the target, between the intersecting point of the vertical and horizontal target lines, and the boresight point of the reticles shall not exceed the tolerances specified.

4.3.8.3 Reticle travel.

4.3.8.3.1 Elevation reticle.- Superimpose the zero graduation of the lateral lead scale on the image of the vertical target line and the horizontal reticle line on the image of the horizontal target line. Depress the horizontal reticle line 47.3 mils plus or minus 0.5 mil, as measured on the target, then return the reticle line to the original position. During this excursion, the zero graduation of the lead scale shall not deviate from the image of the vertical target line by more than the tolerance specified.

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4.3.8.3.2 Ballistic reticle.- Position the zero graduation of the lateral lead scale as specified in 4.3.8.3.1. Position the upper extremity of the drift line (boresight point) coincident to the image of the horizontal target line. Traverse the zero graduation 20 mils right and then 20 mils left from the vertical target line. During this excursion, the upper extremity of the drift line shall not deviate from the image of the horizontal target line by more than the tolerance specified.

4.3.8.4 Reticle jump.

4.3.8.4.1 Ballistic reticle.- Position the lines of the reticles as specified in 4.3.8.1. Rotate the azimuth reticle assembly knob to the clockwise stop, then to the counterclockwise stop, then return the vertical reticle line to the zero lead graduation. During this excursion, the boresight point shall not shift or jump perpendicular to the line of travel, at any point, more than the tolerance specified.

4.3.8.4.2 Elevation reticle.- Position the lines of the reticles as specified in 4.3.8.1. Rotate the elevation reticle assembly knob to the clockwise stop, then to the counterclockwise stop, then return the horizontal reticle line to the boresight point. During this excursion, the zero lead graduation shall not shift or jump perpendicular to the line of travel, at any point, more than the tolerance specified.

4.3.8.5 Reticle accuracy.- Position the boresight point of the ballistic reticle coincident with the image of the horizontal target line. The angular subtense between the boresight point (zero range) and the specified range line, as measured on the target, shall not exceed the limits specified. This test shall be conducted with the telescope adjusted for minimum magnification and maximum magnification.

4.3.8.6 Reticle alinement.

4.3.8.6.1 Elevation reticle.- Position the lines of the reticles as specified in 4.3.8.1. Adjust the horizontal reticle line, if necessary, so that one extremity is coincident to the horizontal target line. The other extremity of the reticle line shall not deviate from the horizontal target line by more than the tolerance specified.

4.3.8.6.2 Ballistic reticle.- Position the lines of the reticles as specified in 4.3.8.1. Adjust the vertical line, if necessary, so that one extremity is coincident to the vertical target line. The other extremity of the reticle line shall not deviate from the vertical target line by more than the tolerance specified.

4.3.9 Reticle illumination.- The reticle pattern shall be legible when illuminated and viewed in a darkened area. In the absence of an Instrument Light,M36, a device consisting of two (2) 1-1/2 volt batteries with a 0.35 to 0.5 amp bulb shall be used to illuminate the reticle. This device shall be properly masked to direct the beam of light into Window (Red) A7682997.

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4.3.10 Torque.

4.3.10.1 Power ring torque (+60°F to +90°F).- This test shall be conducted with the aid of a device, special testing, affixed to the power ring. When the power ring is rotated by means of the device, from clockwise stop to counterclockwise stop then returned to the clockwise stop, the recorded torque shall not exceed the value specified for room temperature.

4.3.10.2 Diopter ring torque (+60°F to +90°F).- This test shall be conducted with the aid of a device, special testing, affixed to the diopter ring. When the diopter ring is rotated by means of the device, from clockwise stop to counterclockwise stop then returned to the clockwise stop, the recorded torque shall not exceed the value specified for room temperature.

4.3.10.3 Reticle adjustment (+60°F to 90°F).- This test shall be conducted with the aid of a device, special testing, affixed to the reticle assembly knob. When the reticle knob is rotated by means of the device, from clockwise stop to counterclockwise stop then returned to the clockwise stop, the recorded torque shall not exceed the value specified for +60°F to + 90°F. This test shall be performed on the azimuth and elevation reticle assembly knobs.

4.3.11 Leak test.- The telescope shall be fully encased in a chamber and pressure tested to detect for leakage. This test shall be conducted by use of a device, special testing, that shall be capable of applying the specified pressure to the exterior surfaces of the telescope for the time period specified. This device shall also have a means of measuring pressure drop within the tolerance specified. Remove the telescope rubber eyeshield and seal the vent hole in the compensating bellows chamber with pressure sensitive tape, or equal, for this test. Any leakage of the telescope shall not exceed the psig tolerance specified.

4.3.12 Environmental test.- The contractor's Government approved refrigeration and heating test chambers shall be used to perform the environmental test.

4.3.12.1 Storage temperature.- Each telescope in the sample selected for environmental testing shall be subjected to the specified ambient temperatures for the time period indicated. Caution must be exercised not to expose the telescope to thermal shock during the environmental test. Temperature changes shall be gradual.

4.3.12.2 Physical failure, post environmental.- Upon return to room temperature (+60°F to + 90°F), subsequent to exposure to each temperature extreme, the telescope shall be subjected to a visual and tactile examination for internal and external damage to mechanical and optical components and deterioration of eyeshield.

4.3.12.3 Operating temperatures.- This test may be conducted concurrent with the storage temperature test.

4.3.12.3.1 Power ring torque, diopter ring torque and reticle adjustment (-40°F and + 150°F).- When thermal equilibrium of the telescope is reached at minus 40°F and plus 150°F, respectively, the torque required to rotate the power ring, diopter ring and reticle adjustment knobs shall not exceed the torques specified for these temperature extremes. These tests shall be conducted as specified in 4.3.10.

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4.3.13 Post environmental performance test.- Each telescope that has been subjected to the environmental tests (storage and operating) shall be subjected to the tests specified in 4.3.1 to 4.3.11 inclusive, and shall meet the requirements specified.

4.3.14 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

4.3.15 Accuracy requirements for devices, special testing.

4.3.15.1 Device to test vibration requirements.- The device shall be a type capable of producing vertical motion at a constant frequency of thirty cycles per second with an amplitude of 1/6 inch (1/8 inch maximum excursion). This device shall be equipped with an adapter which will permit the mounting of the telescope to the adapter in a normal operating position. The base of the adapter shall be secured firmly to the device.

4.3.15.2 Device to test parallax, collimation, reticle travel, reticel jump, reticle accuracy and reticle alinement.- The device shall be equipped with a suitable means to permit mounting the telescope in a horizontal position with the locating pin on the rear collar vertical with 0.1 mil. The axis of the spherical bearing surfaces of the mounting collars establishes the geometric axis of the telescope, therefore, the device shall be so constructed to utilize these bearing surfaces for mounting the telescope in the device.

4.3.15.2.1 An appropriate target to be utilized for measuring the projected telescope reticle pattern with acceptance and rejection limitation markings or pips marked thereon, the angular tolerance of which shall not exceed 0.1 mil, shall be provided. The target shall be utilized for testing parallax, collimation, reticle travel, reticle jump, reticle accuracy and reticle alinement.

4.3.15.3 Device to test leak test requirements.- The device shall be a type capable of applying 10 psi ±2 psi external pressure to the exterior surfaces of the telescope. This device shall be equipped with an air-tight adapter which will accommodate the telescope. The adapter shall be connected to the device by means of air-tight connections. The device and adapter shall indicate no loss of pressure when fully charged during a minimum time period of two minutes, as read on a pressure drop type measuring gage. This gage shall be capable of indicating pressure drops in increments of 0.1 psi.

4.3.15.4 Device to test torque requirements.- A torque wrench type tester with a minimum range of from zero to twenty pound inches, the accuracy of which shall be plus or minus 1/4 pound inches throughout the entire zone of excursion, shall be utilized. This device shall be equipped with suitable adapters to accommodate testing of the power ring, diopter ring, and reticle adjustment knobs.

4.3.15.5 Set-up and calibration equipment.

4.3.15.5.1 All set-up and/or calibration equipment necessary for certification of device accuracy shall be furnished and will be come a part of the device.

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4.3.15.5.2 Rigidity, stability, ease of maintenance, and adaptability of set-up and/or calibration equipment with complete provisions are also prime requirements.

4.3.16 Final acceptance.- Final acceptance of a lot of telescopes shall be withheld by the Government inspector until the sample telescopes, representative of that production lot, have satisfactorily passed the applicable requirements of this specification.

4.4 Packaging, packing and marking.- Quality assurance provisions for packaging, packing and marking of the assembled Telescope, M91C shall comply with Specification MIL-T-45070.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-T-45070.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title number and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable stock numbers
(d) Provisions for production model including date of delivery (see 3.1)

6.2 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.3 Definitions.

6.3.1 Device special testing (DST).- Special testing devices are designed to check function of an assembled unit.

6.3.2 Standard measuring equipment (SME).- Standard measuring equipment includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items) and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.3.3 Tactile.- Skillful manipulating or sensing.

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